

INTERNATIONAL STANDARD



**Radio-frequency connectors –
Part 47: Sectional specification for radio-frequency coaxial connectors with
clamp coupling, typically for use in 75 Ω cable networks (type F-Quick)**



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**Radio-frequency connectors –
Part 47: Sectional specification for radio-frequency coaxial connectors with
clamp coupling, typically for use in 75 Ω cable networks (type F-Quick)**

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INTERNATIONAL ELECTROTECHNICAL COMMISSION

RADIO-FREQUENCY CONNECTORS –

**Part 47: Sectional specification for radio-frequency coaxial
connectors with clamp coupling, typically for use in
75 Ω cable networks (type F-Quick)**

FOREWORD

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International Standard IEC 61169-47 has been prepared by subcommittee 46F: R.F. and microwave passive components, of IEC technical committee 46: Cables, wires, waveguides, R.F. connectors, R.F. and microwave passive components and accessories.

This second edition cancels and replaces the first edition published in 2012. It constitutes a technical revision.

The main changes are listed below:

- Subclause 3.2 has been updated to better define gauging.
- Table 2 has been updated for insertion and removal forces.
- Clause 4 has been updated to refer to the new edition of IEC 61169-1.

The text of this standard is based on the following documents:

CDV	Report on voting
46F/272/CDV	46F/306/RVC

Full information on the voting for the approval of this standard can be found in the report on voting indicated in the above table.

This publication has been drafted in accordance with the ISO/IEC Directives, Part 2.

A list of all parts of the IEC 61169 series, under the general title: *Radio-frequency connectors*, can be found on the IEC website.

The committee has decided that the contents of this publication will remain unchanged until the stability date indicated on the IEC website under "<http://webstore.iec.ch>" in the data related to the specific publication. At this date, the publication will be

- reconfirmed,
- withdrawn,
- replaced by a revised edition, or
- amended.

A bilingual version of this publication may be issued at a later date.

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RADIO-FREQUENCY CONNECTORS –

Part 47: Sectional specification for radio-frequency coaxial connectors with clamp coupling, typically for use in 75 Ω cable networks (type F-Quick)

1 Scope

This part of IEC 61169, which is a sectional specification (SS), provides information and rules for the preparation of detail specifications (DS) for RF coaxial connectors with clamp coupling, typically for use in 75 Ω cable networks (type F-Quick).

It describes the interface dimensions with gauging information, electrical and mechanical performance including the mandatory tests selected from IEC 61169-1:2013, applicable to all DS relating to type F-Quick connectors.

This specification indicates the recommended performance characteristics to be considered when writing a DS and covers test schedules and inspection requirements.

NOTE This interface is typically used for indoor connections, which are easily disconnected and reconnected. The typical application is for F-type coaxial receiver leads or F-type coaxial patch cables. The interface may also be known as a Push-on connector. It is preferred to use the fixed (screwed) connectors type F according to IEC 61169-24:2009.

2 Normative references

The following documents, in whole or in part, are normatively referenced in this document and are indispensable for its application. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 61169-1:2013, *Radio-frequency connectors – Part 1: Generic specification – General requirements and measuring methods*

IEC 61169-24:2009, *Radio-frequency connectors – Part 24: Sectional specification – Radio frequency coaxial connectors with screw coupling, typically for use in 75 Ω cable networks (type F)*

IEC 62037 (all parts), *Passive RF and microwave devices, intermodulation level measurement*

IEC 62037-3, *Passive RF and microwave devices, intermodulation level measurement – Part 3: Measurement of passive intermodulation in coaxial connectors*

3 Interface dimensions

3.1 Dimensions

3.1.1 Common dimensions

Millimetres are original dimensions.

All un-dimensioned pictorial configurations are for reference purposes only.

Figure 1 and Table 1 depict the dimensions that are common to any F connector and thus indispensable for compatibility. Examples of specific design with their dimensions are given in 3.1.2 to 3.1.4.

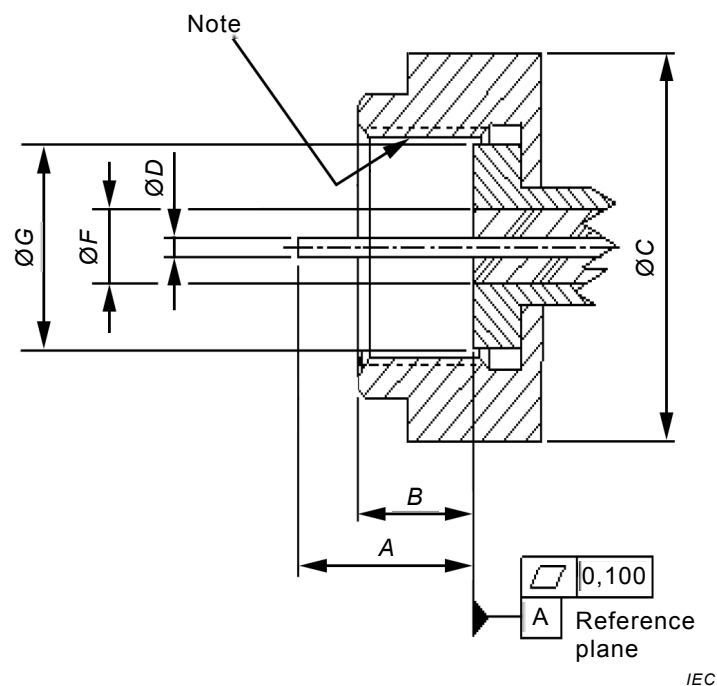


Figure 1 – Connector “F-Quick” type male plug: general dimensions

Table 1 – Connector “F” type male plug (indoor)

Description	Ref.	mm		inches		Additional notes
		Min.	Max.	Min.	Max.	
Inner conductor length	A	6,35	8,63	0,250	0,340	
Length of nut	B	4,00	7,29	0,167	0,287	
Maximum envelope dimension	C		16,61		0,654	
Inner conductor diameter	D	0,64	1,13	0,025	0,044	
Reference plane opening inner diameter	F		6,84		0,230	(1)
Reference plane outer diameter	G	7,11		0,280		
(1) No protrusion of the dielectric beyond the reference plane is permitted.						

3.1.2 Example of connector “F-Quick” type male plug with resilient outer conductor sleeve (indoor) physical dimensions

The connector is shown in Figure 2. Common dimensions are given in 3.1.1.

Dimensions in millimetres

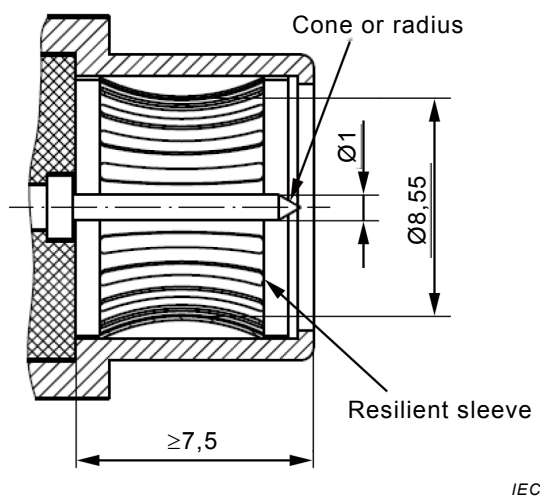


Figure 2 – Example of connector “F-Quick” type male plug with resilient outer conductor sleeve (indoor)

3.1.3 Example of connector “F-Quick” type male plug with slotted outer conductor (indoor) physical dimensions

The connector is shown in Figure 3. Common dimensions are given in 3.1.1.

Dimensions in millimetres

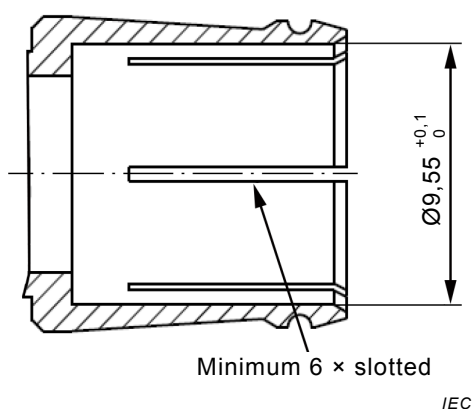


Figure 3 – Example of connector “F-Quick” type male plug with slotted outer conductor (indoor)

3.1.4 Example of connector “F-Quick” type male plug with slotted outer conductor and snap ring (indoor) physical dimensions

The connector is shown in Figure 4. Common dimensions are given in 3.1.1.

Dimensions in millimetres

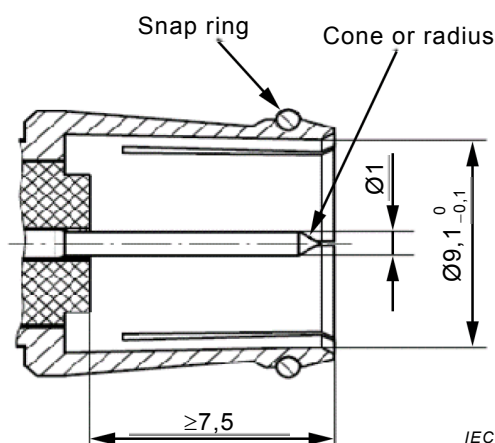
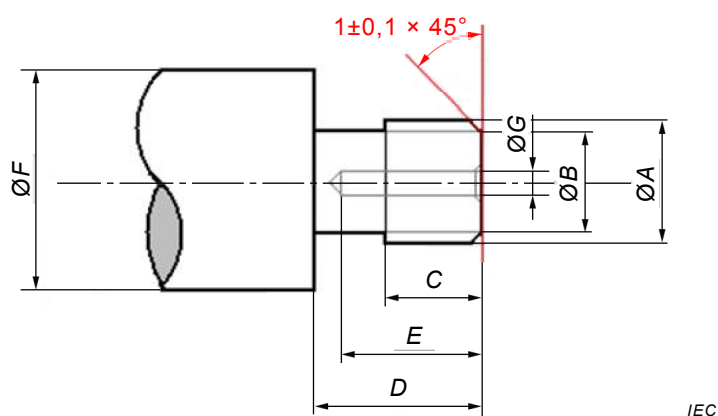


Figure 4 – Example of connector “F-Quick” type male plug with slotted outer conductor and snap ring (indoor)

3.2 Mechanical gauges

Dimensions in millimetres



Material: stainless steel, polish, Ra 0,4-0,1

Description	Dimension	mm	inches
Outside diameter over thread ^a	A	9,35	0,368
Reference plane outer diameter	B	7,35	0,289
Port thread length	C	7,5	0,295
Port length	D	12,69	0,5
Bore depth [min]	E	10,54	0,415
Bulkhead diameter	F	16,6	0,654
Bore diameter [min]	G	3,5	0,14
^a Outside diameter of thread (3/8-32 UNEF 2A).			

Figure 5 – Mechanical gauge for resilient outer conductor

Requirement (see Table 2):

The insertion force required to insert the outer conductor gauge as defined in Figure 5 into the plug shall not exceed 35 N.

The withdrawal force required to withdraw the completely inserted outer conductor gauge from the plug shall be a minimum of 15 N.

4 Quality assessment procedures

4.1 General

Subclauses 4.2 to 4.4 provide recommended ratings, performance and test conditions to be considered when writing a detail specification (DS). They also provide an appropriate schedule of tests with minimum levels of conformance inspection.

4.2 Ratings and characteristics

The r.f. connectors defined in this standard are designed for use with a variety of flexible and semi-rigid coaxial cables and in microwave integrated circuits and similar uncabled applications.

Table 2 – Ratings and characteristics

Rating and characteristics	IEC 61169-1: 2013 Subclause	Values	Remarks, deviations from standard test method
Electrical			
Nominal impedance		$Z_c = 75 \Omega$	Shall meet the requirements of 9.2.1.1 of IEC 61169-1:2013 when terminating a $Z_c = 75 \Omega$ cable
Frequency range			See DS
		5 MHz to 1 GHz 5 MHz to 2 GHz 5 MHz to 3 GHz	For most applications For some satellite applications For some head end applications
Return loss ^a	9.2.1		
– straight styles		> 30 dB up to 1 GHz > 25 dB up to 2 GHz > 20 dB up to 3 GHz	
– right-angle styles ^h		See DS	
– component mounting styles	–	Under consideration	
– solder bucket and PCB mounting styles		Under consideration	
– insertion loss		0,1 dB max. up to 1 GHz 0,2 dB max. at 2 GHz 0,3 dB max. at 3 GHz	(for information only)
Centre contact resistance ^b	9.2.3		
– initial		$\leq 5 \text{ m}\Omega$	
– after conditioning		$\leq 10 \text{ m}\Omega$	
Outer conductor continuity ^b	9.2.3		
– initial		$\leq 2,5 \text{ m}\Omega$	
– after conditioning		$\leq 5 \text{ m}\Omega$	
Insulation resistance ^b	9.2.5		
– initial		$\geq 1 \text{ G}\Omega$	
– after conditioning		$\geq 1 \text{ M}\Omega$	
Proof voltage at sea level ^{c,d}	9.2.6		86 kPa to 106 kPa
– uncabled styles		750 V	
Proof voltage at 4,4 kPa ^{c,d}	9.2.6		4,4 kPa approximately
– uncabled styles			Equivalent to 20 km

Rating and characteristics	IEC 61169-1: 2013 Subclause	Values	Remarks, deviations from standard test method
Environmental test voltage at sea level ^{c,d}	9.2.6		86 kPa to 106 kPa
– uncabled styles		750 V	
Environmental test voltage at 4,4 kPa ^{c,d}	9.2.6		4,4 kPa approximately equivalent to 20 km
– uncabled styles		V	
Screening effectiveness (straight cables only) ^g	9.2.7	$\geq a_s \geq 90$ dB at 1 GHz	$Z_t < 3,2$ m Ω (mated interface)
Intermodulation level	IEC 62037-3	na ^f	
Discharge test (corona effect)	9.2.8	na ^f	Extinction voltage
Centre contact captivation	9.3.5		
– axial force		≥ 20 N	
– torque		See DS	
Engagement and separation	9.3.6		
– Engagement force		35 Nm	
– Separation force		> 15 Nm	
Coupling moment	9.3.6		
– normally moment		na ^f	
– moment resistance		na ^f	
Gauge retention force (resilient contacts)	9.3.4		
– centre		10 N to 20 N	
– outer		na ^f	
Technical tests on cable fixing			
– cable rotation (nutaton)	9.3.7	See DS	
– cable pulling	9.3.8	120 N	
– cable bending	9.3.9	See DS	
– cable torsion	9.3.10	0,1 Nm	
Tensile strength of coupling mechanism	9.3.11	≥ 300 N	
Safety wire hole pull-out bending moment (and shearing force)	9.3.12	> 2 Nm	
Vibration	9.3.3	98 m/s ² 10 Hz to 500 Hz	10 g _n acceleration
Climatic category		A:40/70/21	
Sealing non-hermetic	9.4.7	na ^f	100 kPa to 110 kPa pressure differential
Hermetic	9.4.8	na ^f	100 kPa to 110 kPa pressure differential
Salt mist	9.4.10	48 h	Duration of spraying
Endurance			
Mechanical	9.3.15	See DS for the number of cycles	
High temperature ^e	9.4.5	1 000 h at 70 °C	
^a These values apply to basic connectors. In practice, these may be influenced by the cable used and			

Rating and characteristics	IEC 61169-1: 2013 Subclause	Values	Remarks, deviations from standard test method
reference should always be made to the actual values given in the detail specification.			
b	Values for a single pair of connectors.		
c	Voltages are r.m.s values of a.c., at 40 Hz to 65 Hz, unless otherwise specified.		
d	Some cables usable with these connectors have ratings lower than the values given here.		
e	For certain connectors, the upper temperature limit is restricted by the cable characteristics. Reference should be made to the relevant cable specification. When semi-rigid cables are used, the upper temperature is limited to 115 °C maximum.		
f	na – not applicable.		
g	When interfaces are fully mated.		
h	The test item of this type of connectors is not recommendable.		

4.3 Environmental characteristics for outdoor sockets

See IEC 61169-24:2009.

4.4 Test schedule and inspection requirements

4.4.1 Acceptance tests

There are no group C tests for levels H and M.

Table 3 describes the acceptance tests to be performed.

Table 3 – Acceptance tests

–	IEC 61169-1:2013 Subclause	Assessment level M (higher)				Period	Assessment level H (lower)			
		Test required	IL	AQL	Test required		IL	AQL	Period	
				%				%		
Group A1					Lot by lot				Lot by lot	
Visual inspection	9.1.1	a	II	1		a	S3	1,5		
Group B1										
Outline dimension	9.1.2	a	S4	0,4		a	S3	4,0		
Mechanical compatibility	9.1.2.2	a	II	1		a	S3	1,5		
Engagement and separation	9.3.6	a	S4	0,4		a	S3	1,5		
Gauge retention (resilient contacts)	9.3.4	ia	II	1		ia	S3	1,5		
Sealing										
non hermetic	9.4.7	ia	II	0,65		ia	S3	1		
hermetic	9.4.8	ia	II	0,015		ia	S3	0,025		
Voltage proof	9.2.6	a	II	0,4		a	II	4,0		
Solderability (d)	9.3.2.2	ia	S4	0,4		ia	S3	4,0		
Insulation resistance	9.2.5	a	S4	0,4		a	S3	4,0		
For the symbols, abbreviations and procedures, see the end of Table 4.										

4.4.2 Periodic tests

Table 4 describes the periodic tests to be performed.

Table 4 – Periodic tests

	IEC 61169-1:2013 Subclause	Assessment level M (higher)				Assessment level H (lower)			
Group D1 (d)			6	1	3 years		3	1	3 years
Solderability – connector assemblies	9.3.2.2	ia				ia			
Resistance to soldering heat	9.3.2.3	ia				ia			
Mechanical tests on cable fixing									
cable rotation (nutating)	9.3.7	na				na			
cable pulling	9.3.8	ia				ia			
cable bending	9.3.9	ia				ia			
cable torsion	9.3.10	ia				ia			
Group D2 (d)			6	1	3 years		3	1	3 years
Contact resistance, outer conductor and screen continuity, centre conductor continuity	9.2.3	a				a			
Vibration	9.3.3	a							
Damp heat, steady state	9.4.3	a				a			
Group D3 (d)			1	1	3 years		1*	1	3 years
Dimensions piece-parts and materials	9.1.2	a				a			
Group D4 (d)			6	1	3 years		3	1	3 years
Mechanical endurance	9.3.15	a				a			
High temperature endurance	9.4.5	a				a			
Discharge test	9.2.8								
Climatic conditioning	9.4	na				na			
Group D5 (d)			6	1	3 years		3	1	3 years
Return loss	9.2.1	a				a			
Screening effectiveness	9.2.7	a				a			
Water immersion	9.4.9	ia				ia			
Group D6 (d)			6	1	3 years		3	1	3 years
Contact captivation	9.3.5	a				a			
Rapid change of temperature	9.4.4	na				na			
Climatic sequence	9.4.2	a				a			
Group D7 (d)			1§		3 years		1§		3 years
Salt mist	9.4.10	a							
a suggested as applicable ia test suggested (if technically applicable) na not applicable IL inspection level AQL acceptable quality level * one set of piece-parts each style and variant, unless using common piece parts # for quality conformance , a total of two failures only permitted for level H and 1 failure only for level M from groups D1 to D7 § Group D7 – number of pairs for each solvent (d) destructive tests – specimens shall not be returned to stock									

4.5 Procedures for the quality conformance

4.5.1 Quality conformance inspection

This shall consist of test group A1 and B1 on a lot-by-lot basis and test group D1 to D7 on a periodic basis.

4.5.2 Quality conformance and its maintenance – General procedure

This shall consist of three consecutive lots passing test groups A1 and B1 followed by selection of specimens from the lots as appropriate. These specimens shall successfully pass the specified periodic D tests.

5 Instructions for preparation of detail specifications

5.1 General

Detail specifications (DS) shall use the appropriate blank detail specification (BDS). The following pages comprise the pro-forma BDS dedicated for use with type F-QUICK connectors. As such, it will already have entered on it information relating to:

- a) the basic specification number applicable to all the detail specifications covering connector styles of the type covered by the sectional specification;
- b) the connector series designation.

The specification writer should enter the details relating to the connector style/variant(s) to be covered as indicated. The numbers in brackets on the BDS correspond to the following indications which shall be given.

5.2 Identification of the component

(1) Enter the following details:

- Style: The style designation of the connector including type of fixing and sealing, if applicable;
- Attachment: By deletion of the inapplicable options of cable/wire given for centre and outer conductors;
- Special features and markings: As applicable.

(2) Enter details of assessment level and the climatic category.

- (3) A reproduction of the outline drawing and details of the panel piercing (if applicable). It shall provide the maximum envelope dimensions, also the position of the reference plane and, in the case of a fixed connector, the position of the mounting plane both relative to the front face of the connector.
- (4) Any maximum panel thickness limitations for fixed connectors shall be stated.
- (5) Particulars of all variants covered by the DS. As appropriate, the information shall include:
 - cable types (or sizes) applicable to each variant;
 - alternative plated or protective finishes;
 - details of alternative mounting flanges having either tapped or plain mounting holes;
 - details of alternative solder spills or solder buckets including, when applicable, those for use with microwave integrated circuit (MIC) components.

5.3 Performance

- (6) Performance data listing the most important characteristics of the connector in accordance with the requirements of the relevant sectional specification. Deviations from the minimum requirements shall be clearly indicated. Non-applicable parameters shall be marked “na”.

5.4 Marking, ordering information and related matters

- (7) Insert marking and ordering information as appropriate, together with details of related documents and any invoked structural similarity.

5.5 Selection of tests, test conditions and severities

- (8) “na” shall be used to indicate non-applicable tests. All tests marked “a” by the detail specification writer shall be mandatory.

When using the normal procedure with a dedicated BDS, the letter “a” – for applicable – shall be entered in the “test required” column against each of the tests indicated as being mandatory in the test schedule of the relevant sectional specification. Any additional test required at the discretion of the specification writer shall also be indicated by an “a”.

The specification writer shall also indicate, when necessary, details of deviations from the standard test methods and conditions including any relevant deviations given in the test schedule of the sectional specification.

5.6 Blank detail specification pro-forma for type F-QUICK connector

The following pages contain the complete BDS pro-forma.

(1)		Page 1 of (2)			
ELECTRONIC COMPONENT OF ASSESSED QUALITY IN ACCORDANCE WITH GENERIC SPECIFICATION IEC 61169-1 NATIONAL REFERENCE		(3) . (4) .			
(5) Detail specification for radio frequency coaxial connector of assessed quality			Type		
Style		Special features and markings			
Method of cable/wire+ attachment		centre conductor – solder/crimp+ outer conductor – solder/clamp/crimp + + delete as appropriate			
(6) Assessment level.	Characteristic impedance 75 Ω		Climatic category..55/125./21./		
(7) Outline and maximum dimensions		Panel piercing and mounting details			
(8) Variants					
Variant No.	Description of variant	61196 IEC			
.	.	.	.	.	.
.	.	.	.	.	.
Information about manufacturers who have components qualified under the IECQ Conformity Assessment System is available through the IECQ on-line certificate system.					

(9) Performance (including limiting conditions of use)

Ratings and characteristics	Variant No. Designation	IEC 61169-1:2013 Subclause	Value	Remarks, including any deviations from standard test methods
Electrical				
Nominal impedance			75Ω	
Frequency range Reflection factor		9.2.1	0 GHz to 3 GHz GHz GHz GHz	Measurement frequency range
Centre contact resistance		9.2.3	≤ mΩ ≤ mΩ	Initial After conditioning
Centre conductor continuity		9.2.3	≤ mΩ ≤ mΩ ≤ mΩ ≤ mΩ	Resistance change due to conditioning
Outer contact continuity		9.2.3	≤ mΩ ≤ mΩ	Initial After conditioning
Insulation resistance		9.2.5	≥ GΩ ≥ GΩ	Initial After conditioning
+ Proof voltage at sea level		9.2.6	kV kV kV kV	86 kPa to 106 kPa
+ Proof voltage at 4,4 kPa		9.2.6	kV kV kV kV	kPa (if not 4,4 kPa)
Screening effectiveness		9.2.7	GHz dB at	Z _t ≤.. mΩ
Discharge test (corona) at sea level		9.2.8	≥ V ≥ V ≥ V ≥ V	Extinction voltage
ADDITIONAL ELECTRICAL CHARACTERISTICS				
Intermodulation level		IEC 62037	GHz dB at	Under 2 carries of +43 dBm
+ Voltage values are r.m.s. values at 50 Hz to 60 Hz, unless otherwise specified.				

Ratings and characteristics		IEC 61169-1:2013 Subclause	Value	Remarks, including any deviations from standard test methods
Mechanical				
Soldering – bit size		9.3.2		
Gauge retention resilient contacts – inner contact – outer contact		9.3.4	N N	See Figure 7 and Table 3 See Figure 8 and Table 4
Centre contact captivation – axial force – permitted displacement in each direction – torque		9.3.5	N mm Nm	
Engagement and separation – axial force		9.3.6		Achievable by hand
Strength of coupling mechanism		9.3.11	N	
Effectiveness of cable fixing against – cable rotation		9.3.7	Rotations	
– cable pulling		9.3.8	N N N N	Point of application and duration mm s mm s mm s mm s
– cable bending		9.3.9	cycles cycles cycles cycles	Length of cable and mass mm mm mm mm
– cable torsion		9.3.10	Nm Nm Nm Nm	Duration of applied torque s s s s
Bending moment		9.3.12	Nm	Relative to reference plane
Bumps total		9.3.13	m/s ² to Hz	(g _n acceleration)
Vibration		9.3.3	m/s ² to Hz	(g _n acceleration)
Shock		9.3.14	m/s ² Shape ms	(g _n acceleration)

Ratings and characteristics		IEC 61169-1:2013 Subclause	Value	Remarks, including any deviations from standard test methods
ADDITIONAL MECHANICAL CHARACTERISTICS				
Environmental				
Climatic category				
Sealing non-hermetically sealed connectors		9.4.7	cm ³ /h	100 kPa to 110 kPa pressure differential
Sealing hermetically sealed connectors		9.4.8	10–5 bar/cm ³ /h	100 kPa to 110 kPa pressure differential
Water immersion		9.4.9		
Salt mist		9.4.10	h	Duration of spraying
Additional Environmental characteristics				
<i>ENDURANCE</i>				
Mechanical		9.3.15	operations	
High temperature		9.4.5	at h °C	
ADDITIONAL ENDURANCE CHARACTERISTICS				
CHEMICAL CONTAMINATION				
Resistance to solvents and contaminating fluids to be used. Applicable fluids		9.4.11		
Sulphur dioxide		9.4.12	days	

(10) Supplementary information

– Marking of the component: in accordance with 11.1 of IEC 61169-1:2013 in the following order of procedure 1) Identity of manufacture 2) Manufacturing date code year /week Component identification variant No./designation Identification		
– Marking and contents of package: in accordance with 11.2 of IEC 61169-1:2013 1) Information prescribed in 11.1 of IEC 61169-1:2013 detailed above 2) Nominal characteristic impedance Ω 3) Assessment level code letter 4) Any additional marking required		
– Ordering information: 1) Number of the detail specification/variant code 2) Assessment level code letter 3) Body finish (if more than one listed) 4) Any additional information or special requirements		
– Related documents (if not included in IEC 61169-1:2013 or sectional specification): . .		
– Structural similarity in accordance with 10.2.2 of IEC 61169-1:2013		
Relevant information on a basic style should be entered as variant 01.		

6 Marking

6.1 Marking of component

Each component shall be legibly and durably marked, where space permits and in the following order of precedence, with:

- a) identity code of the manufacturer;
- b) manufacturer's connector identification code or IEC connector designation.

6.2 Marking and contents of package

The package shall be marked with the information prescribed in 6.1 and, in addition, the following information shall be given:

- nominal characteristic impedance;
- manufacturing date code;

c) any additional marking required by the relevant specification.

When required by the relevant specification, the package shall also include instructions for assembling the connector(s) and instructions for the use of any special tools or materials, as necessary.

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